Advanced Embedded Software with STM32, FreeRTOS & Modbus

Hardware Requirements

Develop an Indoor Air Quality Sensor Project Using STM32, FreeRTOS & Modbus and Register-Level Drivers Using CMSIS

Course Hardware Requirements

- STM32F446RE Nucleo Board
- FRAM Breakout Board (MB85RS64V Fujitsu)
- SHT30 Temperature & Humidity Sensor
- SGP40 Indoor Air Quality Sensor for VOC Measurements

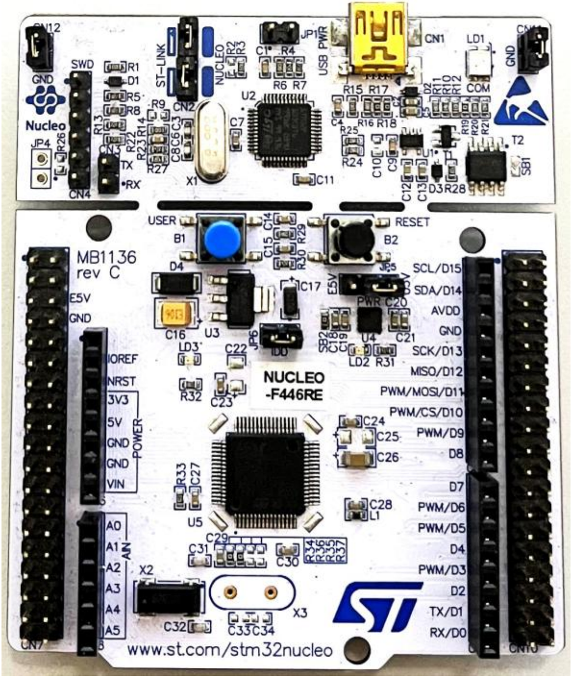
STM32F446RE Nucleo Board

Introduction to STM32F446RE Nucleo Board

- Overview: NUCLEO-F446RE
- Microcontroller Features: STM32F446RE
- NUCLEO Features: Nucleo Board
- Pinout & Connectivity Options: Nucleo Pinout

STM32F4 Series Overview & MCU Specifics

- Overview & Comparison: STM32F4 Series
- STM32F446 MCU Specifics: STM32F446 MCU



STM32F446RE Nucleo Board

Documentation & Resources

- User Manual: NUCLEO-F446RE Documentation
- Community Forums: ST Community
- Tutorials/Resources: ST Step-by-Step, DEEPBLUE MBEDDED, Controllers Tech

Where to Purchase & Cost

- Vendors: DigiKey, Mouser, Amazon

FRAM Breakout Board (MB85RS64V Fujitsu)

Introduction to FRAM Technology

- Basics, Key Features, Applications: Fujitsu FRAM Factsheet

Technical Specifications of the MB85RS64V

- Memory Capacity: 64 Kbit (8K × 8) Bit
- Address Space: 16 bit
- Speed: 20 MHz (Max)
- Communication Interface: Serial Peripheral Interface (SPI)
- Operating Current: 1.5 mA (typical at 20 MHz)

FRAM Breakout Board (MB85RS64V Fujitsu)

Integration with the STM32F446RE Nucleo Board → SPI

- SPI Non-Volatile FRAM Breakout Board - MB85RS64V Fujitsu 64Kbit / 8KByte

Where to Purchase & Cost

- Vendors: Adafruit, DigiKey

SHT30 Temperature & Humidity Sensor

Overview of Sensirion's SHT3x Series

- Basics, Key Features: Sensirion SHT3x Datasheet

Technical Specifications of the SHT30

- Temperature Measurement Range: -40-125°C
- Temperature Measurement Resolution: 0.1°C
- Temperature Measurement Accuracy: 0.3°C

- Humidity Measurement Range: 0-100% RH
- Humidity Measurement Resolution: 0.1% RH
- Humidity Measurement Accuracy: +3% RH

- Operating Current: 1.5 mA

SHT30 Temperature & Humidity Sensor

Integration with the STM32F446RE Nucleo Board → I2C

- I2C SHT3x Temperature And Humidity Sensor

Where to Purchase & Cost

- Vendors: Adafruit, My Local Vendor

SGP40 Indoor Air Quality Sensor

Overview of Sensirion's SGP40 Digital Gas Sensor

- Basics, Key Features: Sensirion SGP40 Datasheet

Technical Specifications of the SGP40

- VOC Index Range: 0-500
- Measuring Range: 0-1000 ppm of Ethanol Equivalent (for sensing various VOCs)

- Preheat Time: Less than 10 Seconds for Initial Startup
- Response Time: 2-3 Seconds to Changes in VOC Levels

- Operating Temperature Range: -10℃ to 50℃
- Operating Humidity Range: 0% to 90% RH non-condensing
- Operating Current: 2.6 mA

SGP40 Indoor Air Quality Sensor

Integration with the STM32F446RE Nucleo Board → I2C

- I2C SGP40 Air Quality Sensor Breakout Board - VOC Index

Where to Purchase & Cost

- Vendors: Adafruit

Next: Course Structure